

Prediction of Diabetes Using Machine Learning: A Comparative Study of Logistic Regression, Random Forest, XGBoost, and Custom Models

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Abstract—This project focuses on predicting the likelihood of diabetes based on patient data using machine learning techniques. Traditional models such as Logistic Regression, Random Forest, and XGBoost are compared, along with custom-built models, including a self-defined Logistic Regression and Neural Network (NN) classifier. Performance is analyzed using various evaluation metrics to identify the most effective model for this classification task. The project demonstrates the application of Python tools in data science and machine learning, leveraging feature selection and model tuning techniques to optimize results.

Index Terms—Diabetes prediction, Logistic Regression, Random Forest, XGBoost, Neural Network, feature selection, model evaluation.

I. INTRODUCTION

A. Background and Goal

Diabetes is a growing global health issue, with millions of individuals affected each year. Early diagnosis is critical for better disease management, making predictive models an important tool for healthcare professionals. Machine learning (ML) techniques provide a promising approach to building accurate models for this purpose. The goal of this project is to compare the performance of multiple machine learning algorithms to predict diabetes based on health-related features. Logistic Regression, Random Forest, and XGBoost are explored, and their results are compared with custom models, including a self-defined Logistic Regression model and a simple Neural Network (NN). Performance is evaluated through metrics such as accuracy, ROC-AUC, and confusion matrix to identify the most effective model.

B. Literature Review

Machine learning techniques are widely used for diabetes prediction, with models such as **logistic regression (LR)**, **random forest (RF)**, **XGBoost** and **neural networks (NN)** showing varied performance. **Logistic regression** is interpretable and effective for linear relationships, but struggles with nonlinearity, where more complex models like **Random Forest** and **XGBoost** excel in capturing intricate patterns [1], [2]. **Random Forest** is robust, especially with unbalanced data, while **XGBoost** improves performance by boosting techniques [3], [4].

Neural Networks can capture complex patterns but often overfit on smaller datasets [5]. Feature selection methods such as **SelectKBest** and **XGBoost's** feature importance enhance model efficiency and accuracy by focusing on relevant features [6]. Despite the advantages of complex models, **Logistic Regression** remains a strong baseline, especially for smaller datasets.

II. METHOD

A. Problem Formulation and Model Description

This project aims to predict diabetes based on features such as glucose levels, BMI, age, and insulin, using binary classification. The Pima Indian dataset (768 samples, 8 features) is employed, with a binary output (1 for diabetes, 0 for no diabetes). The following models from the scit-learn package are used: **Logistic Regression** (linear classifier), **Random Forest** (ensemble of decision trees), **XGBoost** (boosting algorithm), and **Self-Defined Models** (custom Logistic Regression and Neural Networks). The custom Logistic Regression model is implemented using gradient descent, and the Neural Network uses a simple feedforward architecture with a sigmoid activation function. Models are evaluated using accuracy and ROC-AUC to identify the most effective diabetes prediction method.

B. Correlation Heatmap

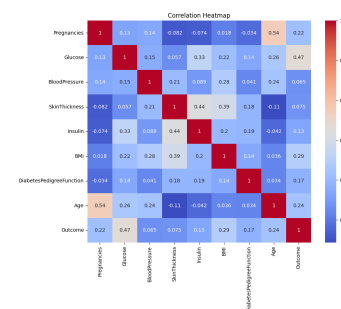


Fig. 1. Correlation Heatmap of Features

The correlation heatmap shows the relationships between various features. High correlations are visible, such as between

Glucose and BMI, are visible, which can help in understanding which features may be more influential in predicting diabetes.

C. Data Preprocessing and Class Imbalance Handling

The dataset contains missing values (zeros), which are replaced with the median of each column. To handle class imbalance, **SMOTE** (Synthetic Minority Oversampling Technique) is applied to balance the class distribution by generating synthetic samples for the minority class, improving model performance on imbalanced datasets.

D. Self-Defined Models and Feature Selection

The custom **Logistic Regression** model is trained using gradient descent with regularization (L1/L2) to reduce overfitting. The model uses a sigmoid activation function to predict diabetes probability. The **Neural Network** consists of one hidden layer, a sigmoid activation function, and is trained using backpropagation and gradient descent. Feature selection is performed using three methods: **SelectKBest** (ANOVA F-test), **Random Forest Feature Importance**, and **XGBoost Feature Importance**. These methods identify the top 5 features that contribute most to the prediction, helping to reduce dimensionality, improve interpretability, and reduce overfitting.

III. RESULTS

A. Base Model Performance

Initially, the **Random Forest (RF)** model performed the best, achieving an accuracy of 83% and an ROC-AUC score of 0.91. These results reflect its strength in handling complex, non-linear relationships within the data. However, after applying **feature selection** (top 5 features), performance slightly decreased across all models.

B. Custom Model Results

The **Self-Defined Logistic Regression** model emerged as the top performer, achieving the highest accuracy of 84% and an ROC-AUC score of 0.9067. The custom implementation allowed for greater control over optimization, contributing to improved performance compared to the built-in Logistic Regression model. In comparison, the **Custom Neural Network** model had an accuracy of 83% and an ROC-AUC score of 0.9104. Although the Neural Network achieved a slightly higher ROC-AUC, its accuracy was slightly lower than Logistic Regression, suggesting that while Neural Networks can capture more complex patterns, they may not always outperform simpler models, especially in smaller datasets where the risk of overfitting is higher.

C. Feature Selection Impact

When feature selection was applied, performance for all models decreased slightly, which was expected as fewer features were available for training. For instance, Logistic Regression with feature selection dropped to an accuracy of 73% and an ROC-AUC score of 0.8151. While feature selection helped improve model interpretability and reduce overfitting,

it also illustrated the trade-off between interpretability and predictive performance. Despite the drop in performance, feature selection provided valuable insights into which features contributed most to the predictions.

TABLE I
MODEL PERFORMANCE RESULTS

Model	Accuracy (%)	ROC-AUC
Logistic Regression (Sklearn)	73	0.8315
Random Forest (Sklearn)	83	0.91
XGBoost (Sklearn)	82	0.8926
Self-Defined Logistic Regression	84	0.9067
Self-Defined Neural Network	83	0.9104
Logistic Regression (Feature Selection)	73	0.8151
Random Forest (Feature Selection)	79	0.8883
XGBoost (Feature Selection)	82	0.8714

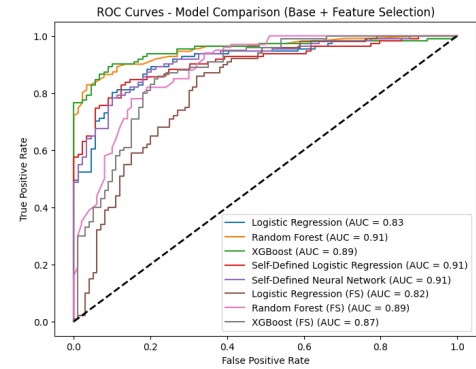


Fig. 2. ROC Curves - Model Comparison (Base + Feature Selection)

IV. CONCLUSION

The analysis demonstrates that Logistic Regression is a robust, efficient, and interpretable model for diabetes prediction, performing comparably to more complex models like Random Forest and XGBoost, especially after feature selection. Although Neural Networks have potential, they require extensive hyperparameter tuning and are prone to overfitting, making them less effective than simpler models for this task.

Limitations and Future Work: This study is limited by its use of a single dataset, and the models' generalizability remains uncertain. The Neural Network could benefit from further optimization. Future work could focus on exploring more advanced regularization techniques, better feature selection methods (e.g., RFE or PCA), and incorporating cross-validation to reduce overfitting and enhance model performance. Additionally, ensemble methods like **LightGBM** or **CatBoost** could further improve prediction accuracy.

In conclusion, Random Forest and Logistic Regression are the most effective models, with Logistic Regression being the most efficient and interpretable. Neural Networks show promise, simpler models are more effective for this task. Future research should focus on optimizing model performance and exploring advanced feature engineering techniques.

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